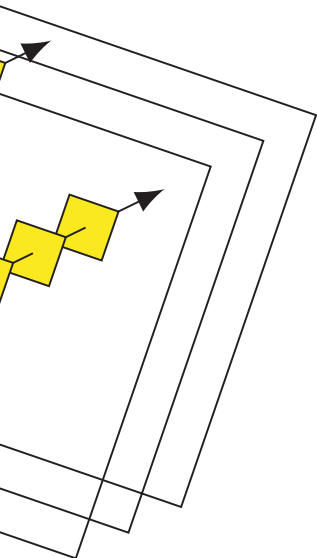
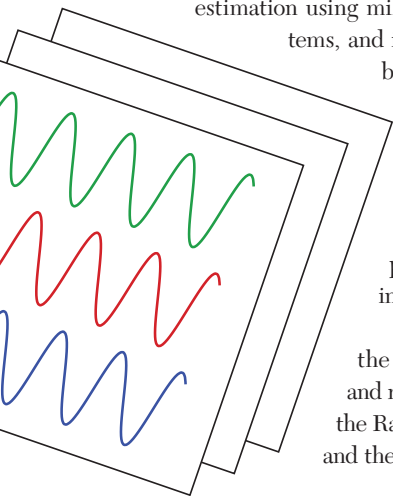
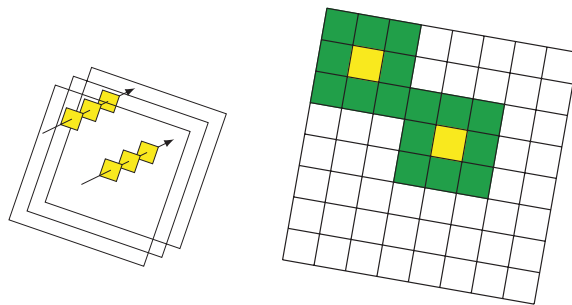




Calibration and Direction-of-Arrival Estimation of Millimeter-Wave Radars

A practical introduction.



Have you read everything about direction-of-arrival estimation in textbooks but are still uncertain how to realize it in practice? This tutorial-like article will help to link the theory with a practical approach for direction-of-arrival estimation using millimeter-wave (mm-wave) radar systems, and it deals with arising challenges. Step by step, it is explained how to move from the measured time domain data to the estimated angular position of the target. Since the target angle estimation is usually carried out after range-velocity processing, the required preprocessing steps are described.

The assessment criteria for forming the relationship between array design and radar performance are expressed with the Rayleigh criterion for angular resolution and the ambiguity function as a measure for

the unambiguous angular range. An in-depth instruction on how to carry out the sensor calibration is provided. Besides the choice of a proper calibration object, the required number of measurements and signal-to-noise ratio (SNR)-related issues are considered and supported by measurement results. A discussion of single- and multitarget measurements with the calibrated sensor concludes this article.

MOTIVATION

In the last decade, mm-wave radars gained a strong interest in the scientific community. The huge advancements in semiconductor technology and the cost-effective development of monolithic microwave integrated circuits for the generation of the high-frequency signal allowed the mass production of radar transceivers. In particular, currently, mm-wave radar sensors are well established as a key sensor technology for many automotive [1], [2], civil [3], [4], and industrial [5], [6] applications. At the present time, it is even possible to simply buy state-of-the-art, off-the-shelf radar platforms for research, educational purposes, or rapid testing [7], [8]. The plug-and-play era of mm-wave radars seems to be just getting started.

The mm-wave radars are able to provide precise distance and velocity measurements even in poor visibility or harsh environmental conditions, where the performances of other sensor technologies start to decrease [1]. Moreover, thanks to the short wavelength, they show appealing compact dimensions. In addition to range and velocity, mm-wave radar sensors typically employ more than one receiving channel to also get the angular position of the target.

In most of the applications, the estimation of the angular position of the target, i.e., direction-of-arrival estimation (DoA), is performed with digital beamforming on the receiver side, since the use of mechanical scanning or phased array systems is too expensive. In particular, the target angle estimation is carried out, usually, after the range–velocity processing, and it is necessary to separate targets that share the same range–velocity bin. Thus, the DoA estimation is rather independent of the radar waveform and modulation scheme, while it is strictly related to the properties of the receiving antenna array.

In recent years, the use of multiple-input, multiple-output (MIMO) radars has become a popular solution to increase the angular resolution of the radar sensor, thanks to the larger virtual aperture that extends the physical aperture of the receiving array. Some examples of mm-wave MIMO radars can be found in [9]–[11].

Processing the signal received by a (virtual) receiving array of antennas for the purpose of angle measurements is a well-established topic in both the antenna and signal processing communities [12]. A number of different approaches have been developed over the last decades; some of them have been summarized, for example, in [12] and [13]. However, the antenna engineer who tackles, for the first time, the array processing with the help of state-of-the-art mm-wave radars could feel overwhelmed when approaching a DoA estimation problem as a newbie. A number of practical questions might come to mind that can rarely be quickly answered by most of the scientific literature.

This article guides the reader toward an efficient DoA estimation using state-of-the-art mm-wave radars. A practical approach is given. Step by step, starting from the very basic preprocessing steps, it explains how to practically perform the DoA estimation and which crucial aspects must be considered. For example, this article tries to explain why the calibration of the sensor plays a significant role and how to carry it out. Moreover, some useful practical hints are given, for example, to save time and processing effort during the calibration and angle estimation phases. No deep radar signal processing prior knowledge is required to follow this tutorial, and it is based on a descriptive rather than formal approach.

THE PREPROCESSING CHAIN

The angular information of a target is present in the reflected signal in the form of the phase relations between the receiving

channels. For the DoA estimation, these phase relations have to be extracted from the received radar signal.

Figure 1 shows a conventional preprocessing chain for calculating and extracting the phase relations of possible targets with a standard multichannel radar using the well-known chirp-sequence modulation [14], i.e., sending multiple very short frequency ramps. The targets manifest themselves as high-amplitude values in certain range–velocity cells, each containing one or more target signals, that share the same range and velocity but have separate angular information.

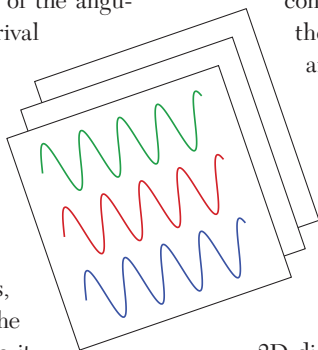
The preprocessing chain starts with the sampled baseband beat signals in the time domain for each ramp on each channel, which—for a single target—manifest themselves as sinusoidal waves. These beat signals undergo—for each channel separately—a series of preprocessing steps before being converted to the range–velocity domain by means of a

2D discrete Fourier transform (DFT), resulting in one range–velocity matrix for each channel. The exact preprocessing steps depend on the respective scenario, but they usually involve some kind of windowing function—for example, a Blackman–Harris window [15]—to suppress unwanted sidelobes introduced by the DFT. Depending on the radar system architecture, additional preprocessing steps in the time domain may be performed, or zero padding can be used to smooth out the resulting range–velocity data.

After this step, possible targets—appearing as high absolute amplitude values in specific range–velocity cells—still have to be extracted from the range–velocity matrix before feeding them into the DoA estimator. This is mostly done to reduce computational effort by discarding cells in the range–velocity domain, which most likely do not contain any targets.

For this reduction, the 3D matrix from the previous steps (range–velocity channel) first has to be accumulated into a single range–velocity matrix, which still includes information from all channels. This can be accomplished by various means, which include the following.

- *Noncoherent integration:* Here, the phase information of the complex amplitudes in the range–velocity cells is discarded, and the absolute values are summed up along the channels [16]. The advantage of this approach is that amplitudes in target cells are contributing equally throughout the channels regardless of their phase relations. The downside, however, is an increase in noise artifacts due to the amplitudes not experiencing any averaging because of the missing phase.
- *Coherent integration:* By summing up the complex amplitude values of the range–velocity cells along the channels, the aforementioned downside of accumulated noise is reduced due to the random nature of the noise process. However, this also leads to the coherent summation of amplitudes for legitimate target cells. This can, potentially, decrease the resulting summed-up amplitude, because the values do not add up ideally due to the naturally occurring phase progression between the channels.



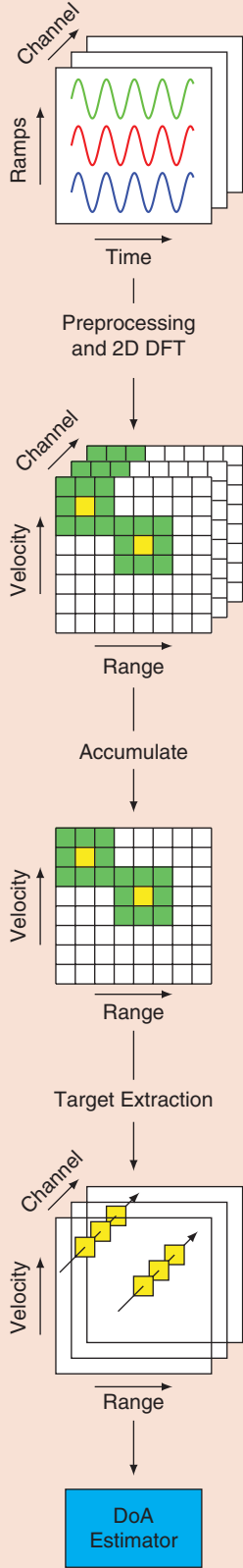


FIGURE 1. The preprocessing chain for extracting the steering vectors of possible targets for a standard multichannel radar using chirp-sequence modulation.

The accumulated matrix then provides the base for the extraction of possible target cells. This usually is accomplished by applying one or more constant false alarm rate (CFAR) algorithms to separate legitimate targets from noise and clutter. Often, an ordered statistics CFAR algorithm, which provides good noise estimation in the presence of multiple targets and clutter edges [17], is deployed either in the range or velocity dimension and, sometimes, in both.

As a final extraction step, a peak search is usually performed in one or two dimensions to find the actual velocity and range of individual target points. This becomes necessary due to the nonidealities of the sampled time signals and the properties of the DFT distorting the ideal frequency peak of a signal.

With the acquired knowledge of the target cells in the range-velocity domain, a cut through the nonaccumulated channel matrices at their respective range-velocity positions is performed. For one target cell, this results in a vector of phase-progressing complex amplitudes describing the steering vector or—in the case of multiple targets in the same range-velocity cell—superimposed steering vectors, which subsequently is fed into the DoA estimator.

THE SIGNAL MODEL

Before going into the details of DoA estimation, it is important to mathematically describe the steering vector obtained at the end of the preprocessing chain sketched in Figure 1. Let M_r be the number of receivers placed in a one-dimensional lattice on the x -axis as in Figure 2. If plane waves impinge from the target, the direction of propagation of the incoming signal, namely, the angle ϑ drawn in Figure 2, is the same for each element of the array. Moreover, only the azimuth plane is of interest in this article. However, it is possible to generalize the DoA estimation toward a 2D problem and to retrieve, thus, both the azimuth and elevation angles of the target. An example of a 2D radar system has been given in [18].

With these assumptions, it is then possible to write the steering vector of the receiving array $\mathbf{y}(\vartheta)$ for the angular direction ϑ as

$$\mathbf{y}(\vartheta) = \begin{bmatrix} e^{j2\pi \frac{x_1}{\lambda_0} \sin \vartheta} \\ \vdots \\ e^{j2\pi \frac{x_{M_r}}{\lambda_0} \sin \vartheta} \end{bmatrix}, \quad (1)$$

where λ_0 is the free-space wavelength, and x_k represents the position of the array element with $k = 1, \dots, M_r$. If the radar

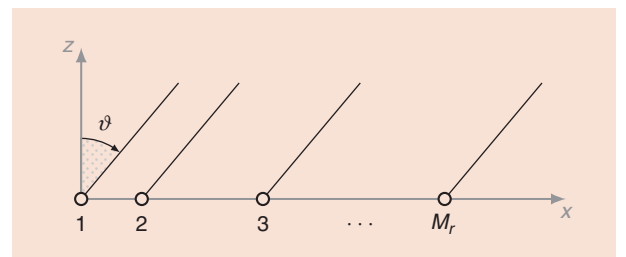


FIGURE 2. The arrangement of the M_r elements of the arbitrary linear array. The angle ϑ is defined within $[-90^\circ, 90^\circ]$.

sensor is a MIMO radar, then, for DoA estimation purposes, it is necessary to consider the steering vector associated with the virtual array, which can be calculated as the Kronecker product of the transmitting and receiving array steering vectors [19] or, equivalently, by applying the vectorization to the product of the transmitting and receiving array steering vectors [20]. The use of a MIMO system does not affect the calibration effort or the implementation of the angular estimation algorithm compared to the non-MIMO case. The study presented in this tutorial applies to both cases. Finally, by considering a number of different directions of interest ϑ_i , it is possible to collect the steering vectors $\mathbf{y}(\vartheta_i)$ to build the steering matrix $\mathbf{Y}$.

THE DETERMINISTIC MAXIMUM LIKELIHOOD APPROACH

This tutorial explains, first of all, the deterministic maximum likelihood approach (DML), where the word *deterministic* comes from the assumption that the received signal waveforms are deterministic and unknown [13]. The DML method becomes quite intuitive and quick to use for a single-target and single-snapshot scenario. Focusing on the single-target case does not narrow the analysis. As shown in the previous section, the DoA estimation takes place after the range–velocity processing. It is, however, quite a reasonable assumption to say that only a single target occupies a range–velocity bin.

The basic idea of the DML approach is to compare the actual measurement results with a reference calibration matrix. This comparison can be mathematically calculated by means of a cross correlation. Indeed, the direction that maximizes the cross correlation, i.e., that returns the highest similarity, gives the estimated DoA ϑ_{est} .

The starting point for the DoA estimation is the calibration of the radar sensor to take into account the typical deviations from the ideal designed antenna system, e.g., due to fabrication errors. Indeed, because of the small wavelength at

mm-wave frequencies, the fabrication inaccuracies can potentially lead to nonnegligible phase errors. This point is also further explained in the “Calibration” section. The results of the reference measurement can be saved into the calibration matrix $\mathbf{C}$ with dimensions $M_r \times N$, where N is the number of samples in the azimuth plane where the calibration measurement has been taken. In other words, the generic element of the matrix c_{kj} is a complex value that describes the collected reference measurement result at the end of the preprocessing chain (see Figure 1) for the angular position ϑ_j and for the k th channel.

More formally, the measurement result for an unknown direction can be saved in the vector $\mathbf{a}$, which is a column vector with dimension M_r . Then, the cross correlation between the measurement and calibration results is given by

$$\mathbf{v}(\vartheta) = \frac{|\mathbf{C}^H(\vartheta) \cdot \mathbf{a}|}{\|\mathbf{C}(\vartheta)\| \|\mathbf{a}\|}, \quad (2)$$

where the symbol $(\cdot)^H$ denotes the complex conjugate (Hermitian) matrix. The dimension of the column vector $\mathbf{v}$ is N .

Finally, the estimated DoA is given by

$$\vartheta_{\text{est}} = \underset{\vartheta}{\operatorname{argmax}} \mathbf{v}(\vartheta), \quad (3)$$

where the function argmax gives the angular position ϑ_{est} at which the cross correlation reaches the maximum (see “Exemplary Direction-of-Arrival Estimation: The Deterministic Maximum Likelihood Approach”).

THE DISCRETE FOURIER TRANSFORM APPROACH

With this simple implementation of the DML procedure presented, it is possible to estimate the position of the target only at the very same points in the azimuth where the calibration results have been collected [21]. Hence, for this basic approach—without any additional processing step—increasing

EXEMPLARY DIRECTION-OF-ARRIVAL ESTIMATION: THE DETERMINISTIC MAXIMUM LIKELIHOOD APPROACH

This example requires the measured calibration matrix $\mathbf{C}$. A MATLAB syntax is chosen to illustrate the required processing steps.

```
% input variables
% a: the measurement steering vector is a
    complex column vector
% C: the calibration matrix with each column
    representing a measurement steering vector

% calculate the cross-correlation of (2)
% -> the Hermitian operator is realized with a com-
    plex conjugation conj and the transposition .'
% -> the normalization is done with the
    Euclidean norm by summing the squared
    values and taking the root of it
```

```
% -> C(:) converts the matrix to a vector,
    hence, a sum of the vector is required
    and not a double sum
A = conj(C) .' * a / (sqrt(sum(abs(a).^2)) *
    sqrt(sum(abs(C(:)).^2)));

% -> convert the spectrum to a logarithmic
    scale
A = 10*log10(abs(A).^2);

% the x axis is dependent on the measured
    angular steps, in this example equal
    calibration steps from [-90,90] degree
    have been captured
plot(linspace(-90,90,numel(A)),A)
```

the number of measurement points during the calibration is beneficial to the purpose of the DoA estimation.

However, for many practical applications and, in particular, for the mass production of mm-wave radar sensors, collecting a large number of measurement points might cause a significant increase in the cost and time dedicated to the calibration of a single sensor. This problem will be become even more clear to the reader after the detailed explanation of the calibration procedure that is presented in the “Calibration” section. Moreover, typically, a large calibration matrix with a sufficient number of measurement points should be saved and then handled by the signal processing unit of the radar system, for example, as a lookup table, and this makes the computational effort of the angle estimation larger. Hence, to reduce the calibration effort, a second approach, which exploits the Fourier transform properties, is introduced here.

This method becomes quite easy to follow by considering a uniform linear array (ULA) as the receiving element. In particular, the similarity to the well-known frequency-domain analysis of time signals must be recalled to easily understand the beamforming and DoA estimation problems. Indeed, it is possible to say that the ULA collects the spatial samples of a certain distribution, of which the Fourier transform—in this case, calculated by means of a DFT—gives the receiving array beampattern [22].

More formally, the spectrum, i.e., the vector $\mathbf{v}(\vartheta)$, can be obtained by collecting the elements relative to the angle ϑ_n :

$$v_n = \sum_{k=0}^{M_r-1} a_k \cdot e^{-j2\pi \frac{n}{M_r} k}, \quad (4)$$

with

$$\vartheta_n \in \arcsin \left[-\frac{1}{2} \frac{\lambda_0}{\Delta x}, \frac{1}{2} \frac{\lambda_0}{\Delta x} \right], \quad (5)$$

where Δx is the constant spacing between adjacent antenna elements in the ULA (see “Exemplary Direction-of-Arrival-Estimation: The Fast Fourier Transform Approach,” or for a more general approach, see “Exemplary Direction-of-Arrival-Estimation: The Discrete Fourier Transform Approach”). The spatial samples a_k contained in the measurement vector $\mathbf{a}$ are described by the index k , which, in turn, represents the k th receiving channel. The index n indicates the spacing of the frequency vector $\mathbf{v}$. The boundaries of (5) are strictly related to the angular range where the DoA estimation can be successfully performed without experiencing any ambiguity. This point is further discussed in the “Assessment Criteria” section. Finally, it is also important to remark that, in v_n , there is no linear dependence on the angle ϑ , but there is on $\sin \vartheta$, similar to (1).

Calculating the DFT of the measurement vector $\mathbf{a}$ is, thus, sufficient to solve the DoA estimation problem. Usually, the Fourier transform is calculated with a reasonable zero padding, which results in a smoother-looking spectrum. The typical reciprocal property of the Fourier transform also applies here, so a wide receiving array generates a beampattern with a narrow main beam, and vice versa. The estimated DoA ϑ_{est} corresponds to the maximum of the DFT, which gives the position of the main beam in the calculated beampattern.

Like the previous approach, a calibration is required for the DFT method, too. The nonideal antenna system must be, indeed, characterized to correctly calculate the DFT and, thus, the beampattern. In particular, the fundamental assumption for using the DFT is that, as reported in (1), for the angular direction of 0° , the associated value in the steering vector should be one for all of the receiving channels. Any deviation from the

EXEMPLARY DIRECTION-OF-ARRIVAL ESTIMATION: THE FAST FOURIER TRANSFORM APPROACH

This example expects a uniform linear array because, then, the discrete Fourier transform equation (4) can be simplified to a fast Fourier transform. Additionally, the calibration—as mentioned in (13)—has to be applied to ensure that a target at 0° leads to equal phases at all channels. A MATLAB syntax is chosen to illustrate the required processing steps.

```
% input variables
% a: the calibrated measurement steering
    vector is a complex column vector

% calculate the fast Fourier transform
% -> the input vector has 32 entries, for a
    smoother spectrum use a zero padding to
    256 values
% -> normalize the spectrum with the number of
    input samples
% -> this is (4) written with the fft function
A = fft(a, 256) ./ numel(a);
```

```
% -> convert the spectrum to a logarithmic scale
A = 10*log10(abs(A).^2);

% as stated in (5), the spectrum is linear in
    sin(theta), hence create the frequency
    vector with arccosin
% -> scale it linearly and then apply the
    arccosin
f = asin( linspace( -0.5 * lambda_0/delta_x, 0.5 *
    lambda_0/delta_x, 256 ) );

% some programs output the spectrum from [0, fs]
    and not from [-fs/2, fs/2], with fs being
    the sampling frequency
% -> the fftshift shifts the spectrum to be
    in [-theta_min, theta_max]
% -> convert the angle from radians to degree
    scaling
plot(rad2deg(f), fftshift(A))
```


ideal 0° value must be compensated. To summarize, before the DFT is calculated, the measurement results must be corrected with the values collected during the calibration for the purpose of obtaining the correct beampattern.

ASSESSMENT CRITERIA

The angular resolution is a fundamental assessment criterion for DoA estimation problems, in particular, to understand whether the antenna system is able to identify two closely spaced targets. From the antenna array theory, it is well known that the resolution properties are directly related to the array aperture. A unique, widely accepted definition of angular resolution is not available in the literature; however, the Rayleigh criterion has often been used as an important reference metric [1]. The Rayleigh criterion relates the angular resolution $\Delta\vartheta$, in degrees, to the array (virtual) aperture L as follows:

$$\Delta\vartheta = \frac{180^\circ}{\pi} 1.22 \frac{\lambda_0}{L}. \quad (6)$$

Thus, as also expected from the properties of the Fourier transform mentioned in the previous section, the larger the aperture L , the better the resolution. Alternatively, the angular resolution of an antenna system can also be defined as a 3-dB or half-power beamwidth. The latter is used in the experimental part of this tutorial as a separation criterion to identify two targets.

An unambiguous estimation of the angular position of the target is another fundamental requirement for successfully solving a DoA estimation problem. In this case, the term *unambiguity* describes the ability of the radar antenna system to uniquely distinguish the DoA of the signal reflected back by the target. This phenomenon comes directly from the well-established concept of grating lobes, which has been nicely explained, for example, in [23] and [24]. Indeed, a grating lobe

EXEMPLARY DIRECTION-OF-ARRIVAL ESTIMATION: THE DISCRETE FOURIER TRANSFORM APPROACH

In contrast to the fast Fourier transform approach, the discrete Fourier transform version can be applied to a nonuniform linear array as well, although this example expects the uniform linear array. Additionally, the calibration—as mentioned in (13)—has to be applied to ensure that a target at 0° leads to equal phases at all channels. A MATLAB syntax is chosen to illustrate the required processing steps. The following example is a practical implementation of (4).

```
% input variables
% a: the calibrated measurement steering vector
% is a complex column vector

% the input measurement steering vector is
% zero-padded to this length
L = 256;

% preallocate the resulting frequency vector
v_n = zeros(L, 1);

% zero-pad the measurement steering vector to
% the same length
a_zp = [a; zeros(L-numel(a),1)];

% calculate the DFT (4) by hand
% -> the measurement steering vector ranges
% from [0,L-1], which is without zero-padding
% [0,Mr-1]
theta = 0:L-1;
% -> this for loop calculates the whole frequency
% vector in contrast to (4) which
% only calculates a single frequency spectrum
% value
```

```
for k = 1:numel(theta)
    % -> the kth frequency spectrum value in
    % (4) consists of a sum, which is here
    % realized with a vector product of a
    % row and a column vector to be more
    % efficient
    % -> the current frequency is theta(k)
    % which is in (4) the n
    % -> the division by Mr in (4) is written
    % as numel(a_zp)
    % -> in the sum, k is increased from
    % [0,Mr-1], which is here [0,L-1] due
    % to the zero-padding. the column vector
    % for the vector product is created
    % with the colon : operator and a
    % transposition .'
    v_n(k) = a_zp.' * exp(-1i*2*pi * theta(k) /
        numel(a_zp) * (0:numel(a_zp)-1)).';
end

% -> convert the spectrum to a logarithmic scale
v_n = 10*log10( abs(v_n).^2 );

% as stated in (5), the spectrum is linear in
% sin(theta), hence create the frequency
% vector with arcsin
% -> scale it linearly and then apply the
% arcsin
f = asin( linspace( -0.5 * lambda_0/delta_x,
    0.5 * lambda_0/delta_x, L ) );
plot(rad2deg(f), v_n)
```

in the receiving array beampattern causes an ambiguity in the angular estimation.

This concept has been formally described in [25], where it is possible to find the definition of the ambiguity function $\chi(\vartheta_i, \vartheta_j)$, here, recalled for the sake of completeness:

$$\chi(\vartheta_i, \vartheta_j) = \frac{|\mathbf{y}^H(\vartheta_i) \cdot \mathbf{y}(\vartheta_j)|}{\|\mathbf{y}(\vartheta_i)\| \|\mathbf{y}(\vartheta_j)\|}. \quad (7)$$

It is the autocorrelation of the receiving array steering vector calculated at the positions ϑ_i and ϑ_j , respectively. An ambiguity occurs when it is not possible to distinguish between the directions ϑ_i and ϑ_j . The autocorrelation should, indeed, return the maximum value of one only if $\vartheta_i = \vartheta_j$ and, ideally, zero in all of the other cases. Usually, the ambiguity function can be depicted as a 3D plot, where the coordinates x and y describe the angles ϑ_i and ϑ_j , while the z -coordinate depicts the magnitude of the function [26], which is always bounded between zero and one.

A simple example of the ambiguity function plot can be seen in Figure 3. The results refer to a ULA of isotropic radiators with six elements spaced λ_0 apart. The highest value of the autocorrelation has been achieved not only along the main diagonal, where $\vartheta_i = \vartheta_j$, but also for some other angular directions, leading,

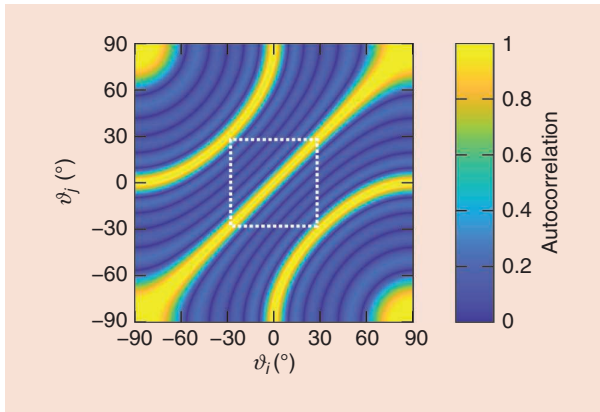


FIGURE 3. The plot of the ambiguity function for an ideal six-element ULA with a spacing equal to λ_0 between the elements. The dotted white area describes the ambiguity-free region where no ambiguities occur.

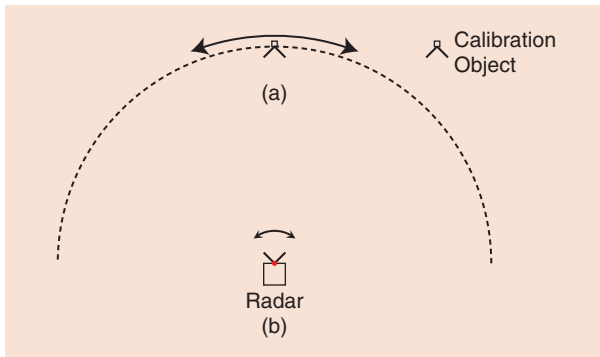


FIGURE 4. A possible calibration setup. Either (a) the calibration object is rotated around the radar or (b) the radar is rotated and the calibration object is located at fixed location.

thus, to ambiguities. The white box at the center of the plot highlights the area where no ambiguities occur. As known from the array theory, to avoid ambiguities in the estimation of the direction ϑ , the following equation must be fulfilled [24]:

$$-\pi \leq 2\pi \frac{\Delta x}{\lambda_0} \sin \vartheta < \pi. \quad (8)$$

In principle, if the field of view (FoV) of the radar is confined to the ambiguity-free region, then it is, ideally, possible to estimate correctly the direction of the target. On the other hand, if the FoV is larger than the ambiguity-free region, then it is possible that some measurements can lead to a wrong DoA estimation.

Avoiding grating lobes is a necessary, but not sufficient, condition for estimating the DoA of a target. For practical applications, it is also important to avoid a high sidelobe level. Indeed, high sidelobes could cover, for example, some weaker targets present in the same range–velocity cell, making, thus, their correct estimation difficult. High sidelobes can also be called *quasi-ambiguities*. Typically, it is possible to adapt the definition of an ambiguity-free region to also take the quasi-ambiguities into account [27]. For example, a threshold for an acceptable sidelobe level can be defined according to the specific application, and, then, the ambiguity-free region is the area where the ambiguity function $\chi(\vartheta_i, \vartheta_j)$ reaches values below that threshold.

CALIBRATION

As already briefly mentioned in the “Signal Model” section, due to differences in the transmission line lengths within the radar systems, manufacturing uncertainties, variances in radio-frequency components, and different power distributions on the transmit channels, there are amplitude and phase variations between the different transmit–receive channels. These effects result in a deviation from the ideal signal model in (1), described in the “Signal Model” section. The signal model in (1) can be adapted to a more realistic one, which is described by

$$\mathbf{y}(\vartheta) = \begin{bmatrix} A_1(\vartheta) e^{j(2\pi \frac{x_1}{\lambda_0} \sin \vartheta + \psi_1)} \\ \vdots \\ A_{Mr}(\vartheta) e^{j(2\pi \frac{x_{Mr}}{\lambda_0} \sin \vartheta + \psi_{Mr})} \end{bmatrix}, \quad (9)$$

with phase offset ψ_k and amplitude A_k for the virtual channel k . Both the amplitude term and the phase are angle dependent. The amplitude term can be neglected if there are no large differences in the amplitude with respect to the incident wave angle. This is usually fulfilled if the antenna is not focusing in the radiation plane—exemplary in the presence of a focusing lens—where the DoA estimation is performed. In the following, the amplitude is neglected.

For the calibration, let us assume a point-like target in the far field of the antenna array with a known azimuth angle $\vartheta = \vartheta_0$. For a plane wave impinging the antenna array, the phase at the receive antenna is described by the exponential term in the signal model of (9). By applying an angle-dependent calibration measurement, the phase term $\psi(\vartheta) = 2\pi(x_k/\lambda_0) \sin \vartheta + \psi_k$ can be determined. This can be achieved by a calibration measurement as a function of the azimuth angle ϑ (see Figure 4).

Basically, there are two calibration setups:

- **Rotating calibration object:** The radar stays fixed, and the calibration object is rotated around the phase center of the radar antenna. This is the most intuitive approach but requires a very accurate positioning system with a large size, which is difficult to realize in practice.
- **Rotating radar:** The same result can be achieved by rotating the radar around its antenna phase center point. This is approximated by a rotation around the center of the antenna front end. This second approach is more accurate and realizable in practice as it can be done using a stepper motor rotating the radar. Thus, this case is considered in this article.

As the calibration is conducted for discrete angular step-sizes, this results in the matrix $\mathbf{C}$ and can be directly applied to the DML of (2). Furthermore, the nonidealities of the antenna array and the phase offsets are included within the calibration measurement.

A consideration of only the phase of the calibration measurement with respect to the first antenna element results in the phase progression curves (see Figure 5). They behave according to the exponential part of the signal model in (1) and appear as straight lines if visualized as a function of $\sin \vartheta$. In the following section, some key aspects of performing a reliable calibration are discussed.

THE MEASUREMENT SYSTEM

An experimental 77-GHz MIMO radar sensor [11] is used to illustrate the different calibration effects and the measurement examples. The time domain data are directly sampled with a sampling frequency of 20 MHz and saved for detailed signal processing later on. The bandwidth is set to 2 GHz, resulting in a range resolution of 7.5 cm. Four transmitters and eight receivers form a ULA with a $\lambda_0/2$ spacing. The virtual aperture has a length of $15.5 \lambda_0$, and, with (6), this leads to an angular resolution of 4.5° . Each single physical antenna element, namely, the transmitters and receivers, is an eight-element, series-fed patch antenna array using the substrate RO3003. To permit the easy replacement of different antenna front ends, the antenna structure is connected to a waveguide transition, simply allowing the use of different antenna setups as shown in Figure 6. To cover a wide angular range in azimuth, the antenna array is focused only in the E-plane and has a single patch characteristic in the H-plane. The calibration and measurement examples are carried out in an anechoic chamber.

THE INFLUENCE OF THE SNR

In the following, the influence of the SNR on the calibration is discussed. The SNR in a measurement is estimated using the mean of all samples not classified as a target by the CFAR and the power level of the current considered target. In Figure 5, the phase progression over the calibration angles is shown. The used calibration objects are corner reflectors, which differ in their radar cross sections (RCSs). For Figure 5(a), a corner reflector with an RCS of 6.7 dBsm is used. This results in a measured SNR value of 32 dB, which is obtained by noncoherent integration of the receive channels. As the value for the

SNR is high, the phase curve is smooth and shows a linear trend over a wide angular range. The phase curve is more noisy for large angles, which results from the choice of the antenna element. For the used antenna element, the considered angular

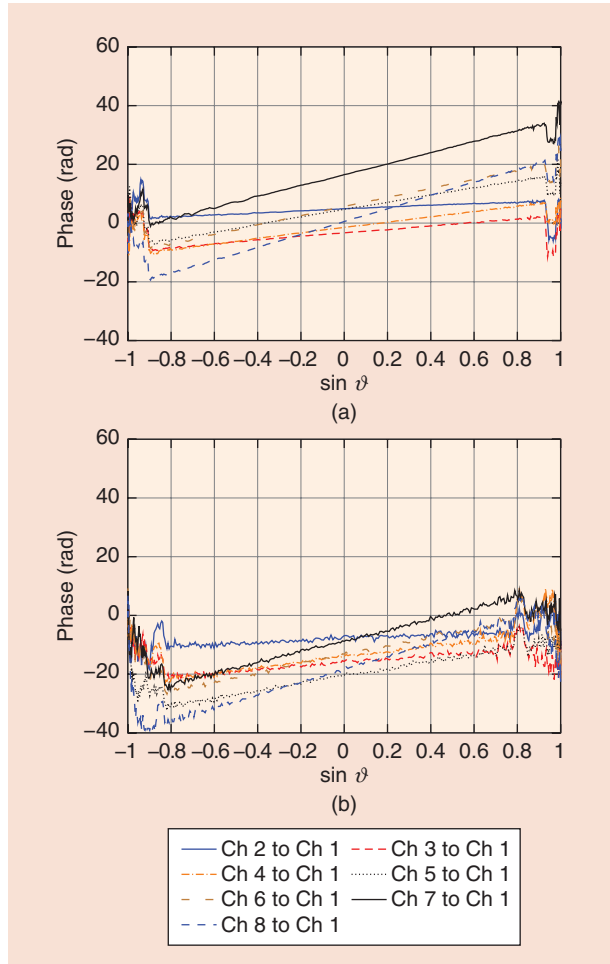


FIGURE 5. The measurement of the phase progression over the calibration angle for some receive channels relative to the first receive channel. (a) A high SNR of 32 dB leads to a smooth line for a wide angular range. (b) A low SNR of 11 dB results in a noisy phase relation and a smaller angular range in which reliable results can be obtained. Ch: channel.

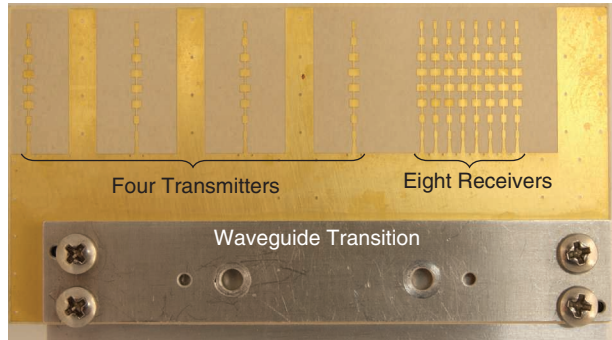


FIGURE 6. The antenna front end with the four transmitters and eight receivers forming a uniform linear virtual array with a $\lambda_0/2$ spacing between adjacent elements.

range can be confined to $\arcsin(\pm 0.9) \approx \pm 64^\circ$, which is the linear region in Figure 5(a).

In contrast, Figure 5(b) shows the same phase plot for a corner reflector with an RCS of -16.4 dBsm and an SNR value of 11 dB. The reduced SNR leads to the phase plot becoming less smooth and to a smaller angular range in which reliable results can be acquired.

THE CHOICE OF CALIBRATION OBJECTS

There are several calibration objects that might be used for the setup in Figure 4. Commonly used calibration objects are corner reflectors, spheres, and cylinders. They are usually fabricated from well-reflecting metals. These objects are most suitable as their RCSs, which describe the amount of reflected power, can be analytically described. Additionally, their RCSs are very constant for a large angular range.

In the following, only the sphere is compared to the corner reflector. For a sphere, assuming that the sphere diameter D is much larger than the wavelength λ_0 , the RCS is the projection of its cross-sectional area. In contrast, the RCS of the corner reflector can be described by

$$\sigma_{\text{corner}} = \frac{4\pi}{3} \left(\frac{a}{\sqrt{2}} \right)^4, \quad (10)$$

with the outer edge length a . For an exemplary RCS = 0.1 m^2 , at 77 GHz, the corner reflector has an outer edge length a of 3.47 cm, whereas a sphere requires a diameter $D = 25$ cm. The RCS of the corner is frequency dependent and increases with frequency.

To conduct a reliable calibration, it is reasonable to place the calibration object in the far field of the antenna. Additionally, the antenna array should also be in the far field of the calibration object. Only for these cases can plane waves be assumed, and (9) is fulfilled. The far-field distance R can be defined according to [28] by

$$R > 2 \frac{L^2}{\lambda_0}. \quad (11)$$

The size L is the largest value between the antenna array size and the size of the calibration object.

In practice, corner reflectors are mostly used as a calibration object in the mm-wave range as it is easier to fulfill the far-field condition in (11). For the exemplary corner reflector with an edge length a of 3.47 cm and the corresponding sphere with a diameter D of 25 cm, the far-field condition is fulfilled at 62 cm for the corner reflector and at 32.7 m for the sphere, which is difficult to realize with sufficient SNR (see the “Influence of the SNR” section).

A LIMITED NUMBER OF ANGULAR STEPS: A FASTER CALIBRATION

The DML estimation can only be applied to discrete azimuth angles for which a calibration exists (see the “Signal Model” section). The reasonable number of steps is determined by the resolution of the array given by (6) and the possible ambiguity-free region defined by (8). To allow a precise DoA estimation, the step-size is set to 0.5° . The ULA spacing is $\lambda_0/2$, which results in an ambiguity-free region of $\pm 90^\circ$. As the antenna consists of single elements, their radiation patterns must also be taken into account. As mentioned for the SNR considerations shown in Figure 5(a), the phase trend is linear in the domain of, roughly, $\sin(\vartheta) \in [-0.9, 0.9]$, leading to a DoA range of $\pm 64^\circ$. Together with the small angular increment, this would result in 257 different calibration measurements to determine the steering matrix $\mathbf{C}$ for the DML estimation.

To reduce the number of required measurement points for the steering matrix $\mathbf{C}$, the key idea is to determine the phase offsets ψ_k of (9) for each single channel and to compensate them. This allows for creation of the steering matrix $\mathbf{Y}$ based on model (1) for a single steering vector without phase deviation. In contrast to the measured matrix $\mathbf{C}$, the created ideal steering matrix $\mathbf{Y}$ does not contain any noise and, therefore, leads to a smoother DoA result. In addition, the angular step-size can be chosen as small as desired without increasing the measurement effort. Moreover, the removal of the phase offsets is, for the application of the Fourier transform-based DoA estimation, also mandatory, as mentioned in the “Signal Model” section.

In theory, only a single calibration measurement for a target at 0° would be sufficient for the determination of the phase offsets ψ_k . As it is affected by noise, an averaging with several measurements should be considered. Instead of repeating the evaluation of a target under 0° , a certain angular range is considered. The phase difference $\psi_{k,1}(\vartheta = 0^\circ)$ for each channel with respect to the reference channel, e.g., the first one, is then determined by linear regression.

The influence of different calibration ranges is depicted in Figure 7. The overall phase trend is ideally approximated with a wide angular range (—), but, also, a reduced range (---) already leads to a good approximation. It must be

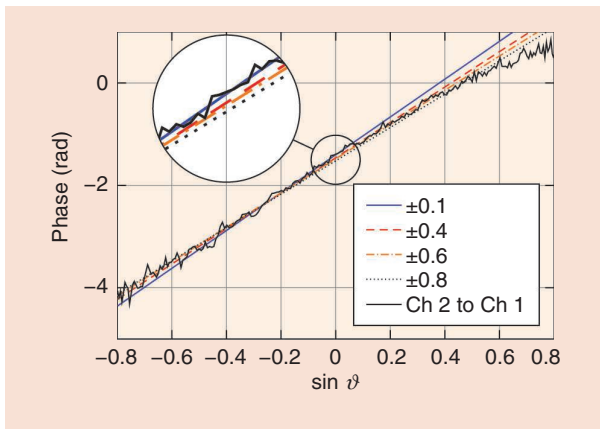


FIGURE 7. Instead of measuring the whole angular range (—), smaller ranges are used for the linear regression and compared. The phase value at 0° is the phase difference $\Delta\psi_{2,1}(\vartheta = 0^\circ)$ between the second and the reference channel.

ensured that the phase trend is a straight line and no phase jumps occur. Therefore, the step-size for this smaller calibration domain is defined so that the maximum relative phase progression between two calibration steps is much smaller than $\pm 180^\circ$ to take phase noise into account. A step-size of 0.5° leads to a phase progression between two calibration angles of

$$\Delta\psi_{\max} = 360^\circ \frac{L}{\lambda_0} (\sin 0.5^\circ - \sin 0^\circ) \approx \pm 48.7^\circ \quad (12)$$

for the maximum progression between the first and last virtual channels. For an insufficient calibration step-size of 2° , the maximum phase progression is $\pm 194.7^\circ$, leading to phase jumps, as can be seen in Figure 8.

With the determined phase differences $\psi_{k,1}(\vartheta = 0^\circ)$, each single channel is corrected in the time or frequency domain with

$$\tilde{s}_k = s_k e^{-j\Delta\psi_{k,1}(\vartheta = 0^\circ)}. \quad (13)$$

After calibration, the Fourier transform-based DoA is possible, as is the creation of the steering matrix $\mathbf{Y}$, which is later referenced as *ML ideal*.

For a non-ULA array, the DML approach is the easiest choice, as the exact element position is not required. In case of the Fourier transform, either the Fourier transform must be adapted to operate with nonequidistant samples or the missing element positions with respect to the ULA should be assumed to be zero.

MEASUREMENT EXAMPLES FOR DoA ESTIMATION

Two measurement scenarios are presented to highlight different aspects of the DoA estimation. For a single-target scenario, the influence of the SNR in the calibration on the DoA estimation is shown. In the case of two close targets, the three different angle estimation approaches are compared.

THE SINGLE-TARGET SCENARIO: INFLUENCE OF THE SNR IN CALIBRATION

The DoA estimation for a single corner reflector under 20° is shown in Figure 9. With the DML estimation (—), the expected DoA is extracted. For angles larger than $|\vartheta| > 64^\circ$, the DoA estimation starts degrading. This behavior is expected, as the phase trend in Figure 5(a) is stable for the same angular domain. This is the limitation of the single antenna element of the evaluated array. The sidelobe level is as expected, at approximately -13 dB, as no window is applied, which coincides with the theory.

The same DoA estimation is, once again, performed with the calibration data of Figure 5(b) with lower SNR. Although the DoA can be extracted, the resulting angular spectrum is noisy (---). The loss in SNR should be avoided, as additional possible weak targets in the same range-velocity cell could not be detected.

The phase trend is interpolated, and the time domain data are calibrated to allow the application of the ideal steering

matrix $\mathbf{Y}$. With this matrix, the DML estimation is, once again, performed (---). The resulting angular spectrum has roughly the same shape as the DML approach based on the measured steering matrix, although it is smoothed and has lower sidelobes.

TWO CLOSE TARGETS: A PERFORMANCE COMPARISON

Two corner reflectors of equal size are placed at the same radial distance with a separation of 5° , which is larger than the theoretical angular resolution of 4.5° . This measurement setup and the experimental radar sensor mounted on a rotary stand are shown in Figure 10. The insert is a close-up of the radar sensor with the antenna front end. The DoA estimation is performed without knowledge or determination of the number of presented targets. Thus, the three different DoA algorithms are applied without any modification, and the

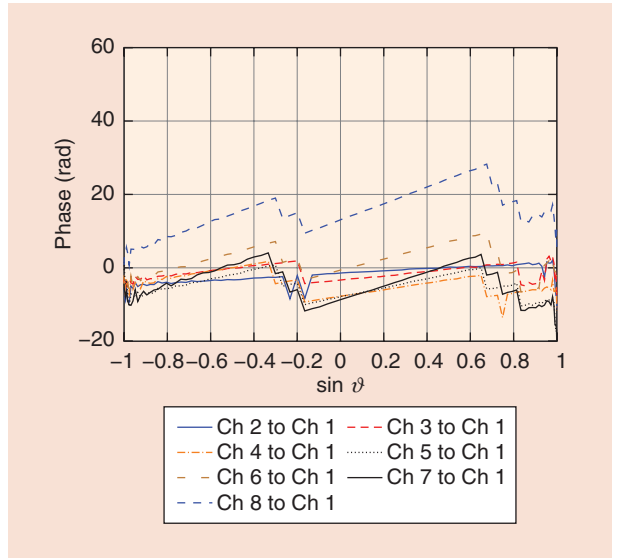


FIGURE 8. Calibration with a too large step-size of 2° leads to phase jumps and prevents a reliable linear regression.

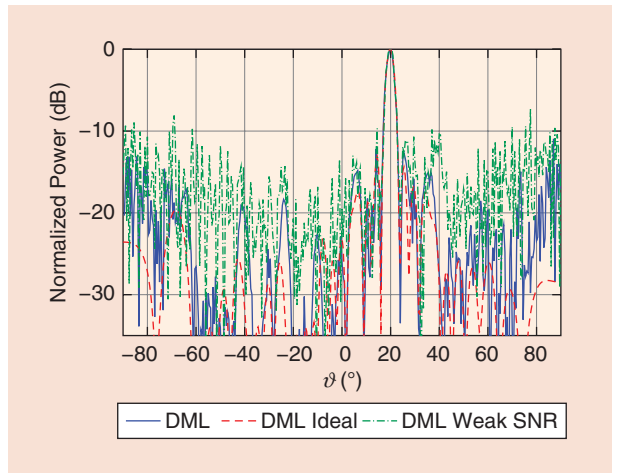


FIGURE 9. The measurement of a single target with the measured steering matrix (—), the idealized matrix (---), and the steering matrix obtained of a weak SNR target (---).

results are depicted in Figure 11 to show that close targets can be distinguished.

The DoA based on the measured steering vector (—) shows, once again, at the edge of the evaluable range, the limitation of the antenna elements. The drop in amplitude between the two targets is much larger than 3 dB, which is considered as the separation criterion.

The exploitation of the idealized steering vector (---) has—as in the single-target case—fewer ripples compared to the normal DML approach. The steering entries for the edge of the evaluable range are calculated based upon (1); hence, they show no artifacts.

The time domain data are calibrated, and the Fourier transform using a zero padding to 256 values is applied for DoA estimation (-.-.-). This approach yields the same result as the idealized steering vector (---). The reason is that the Fourier transform is the cross correlation of the measured steering vector and the complex sinusoidal ideal steering matrix, similar to the DML approach.

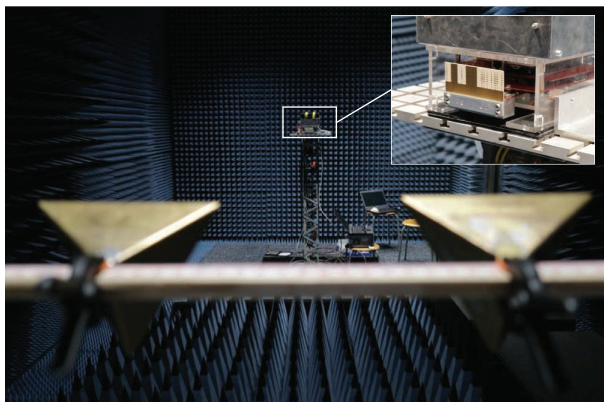


FIGURE 10. The measurement setup showing two close corner reflectors with the same RCS. The experimental radar sensor is mounted on a rotary stand visible in the background and shown as an insert.

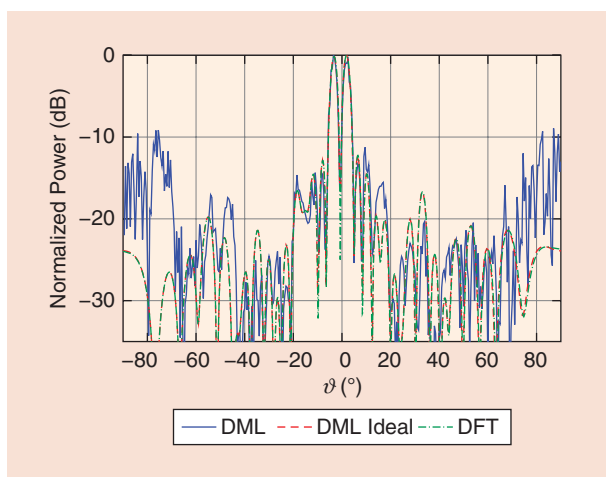


FIGURE 11. The measurement of two close targets with the measured steering matrix (—), the idealized matrix (-.-.-), and the DFT (-.-.-).

CONCLUSIONS

A practical and straightforward tutorial for DoA estimation with state-of-the-art mm-wave radars, supported by measurements on the basis of a 77-GHz MIMO radar sensor is presented. For DoA estimation, the sensor calibration is a key task, and a step-by-step description is provided. Different aspects influencing the quality of the calibration and the reliability of two basic DoA estimation techniques, the DML and DFT approaches, are discussed. Especially, the effect of the SNR and the directly related choice of a proper calibration object on the estimation are shown. Measurements with an array with a theoretical angular resolution of 4.5° are performed. For a scenario with two targets in the same range-velocity bin separated by only 5° , the measurement results show that the targets can clearly be distinguished, and this verifies the presented procedure here.

This article gave a practical overview of the DoA estimation with mm-wave radars. Further reading about additional DoA estimation techniques can be found in [13] and [29], such as the Bartlett beamformer maximizing the power of the DoA; the Capon beamformer minimizing the power contribution by noise; or subspace-based methods, such as multiple signal classification. The ambiguity function discussed here as an assessment criterion has been introduced for single-target scenarios. To extend the ambiguity function toward higher orders, see [30].

When time-division multiplexing is used to achieve orthogonal signals for MIMO radars, special care must be taken in the case of moving targets, which leads to phase errors and can result in inaccurate or unusable angular estimation. Either overlapping elements [31] or an adaption of the DFT [32] can be used to overcome this issue. Last but not least, sparse antennas can be employed to enhance the angular resolution while maintaining a limited number of transmitters and receivers. By using compressed sensing, the missing antenna elements can be reconstructed to mitigate the high sidelobes [33].

ACKNOWLEDGMENTS

This work was partially supported by the German Federal Ministry of Education and Research in the framework of the projects KoRRund and radar4Fad and by the Ministry for Science, Research, and Art Baden-Württemberg within the project Zentren für angewandte Forschung an Hochschulen für angewandte Wissenschaften (ZAFH) MikroSens.

AUTHOR INFORMATION

Claudia Vasanelli (claudia.vasanelli@alumni.uni-ulm.de) received her Ph.D. degree from Ulm University, Germany, in 2019. Her research interests include millimeter-wave antenna systems, in particular, for automotive radar applications. She is a Student Member of the IEEE.

Fabian Roos (roos@ieee.org) is currently pursuing his Ph.D. degree at Ulm University, Germany. His research interests include automotive radar and environmental perception

with a focus on adaptivity for chirp-sequence-modulated automotive radar sensors. He is a Student Member of the IEEE.

André Dürr (andre.duerr@uni-ulm.de) is currently pursuing his Ph.D. degree at Ulm University, Germany. His research interests include high-resolution imaging radars in the upper millimeter-wave range as well as performance degradation of noncoherent multichannel radar systems and their synchronization. He is a Student Member of the IEEE.

Johannes Schlichenmaier (schlichenmaier@ieee.org) is currently pursuing his Ph.D. degree at Ulm University, Germany. His research interests include advanced signal processing for cooperative radar networks as well as the detection and estimation of extended radar objects in automotive scenarios. He is a Student Member of the IEEE.

Philipp Hügl (paul.huegler@uni-ulm.de) is currently pursuing his Ph.D. degree at Ulm University, Germany. His research interests include imaging radar front ends for small unmanned aerial vehicles. He is a Student Member of the IEEE.

Benedikt Meinecke (benedikt.meinecke@uni-ulm.de) is currently pursuing his Ph.D. degree at Ulm University, Germany. His research interests include system concepts for multistatic coherent radar networks and the accompanying signal processing. He is a Student Member of the IEEE.

Maximilian Steiner (maximilian.steiner@uni-ulm.de) is currently pursuing his Ph.D. degree in electrical engineering at Ulm University, Germany. His research interests include system concepts for radar sensor networks in automotive and robotic scenarios as well as the accompanying signal processing. He is a Student Member of the IEEE.

Christian Waldschmidt (christian.waldschmidt@uni-ulm.de) was appointed director of the Institute of Microwave Engineering at Ulm University, Germany, as a full professor in 2013. His research interests include radar, sub-millimeter-wave engineering, and antennas. He is a Senior Member of the IEEE.

REFERENCES

- [1] J. Hasch, E. Topak, R. Schnabel, T. Zwick, R. Weigel, and C. Waldschmidt, "Millimeter-wave technology for automotive radar sensors in the 77GHz frequency band," *IEEE Trans. Microw. Theory Techn.*, vol. 60, no. 3, pp. 845–860, Mar. 2012. doi: 10.1109/TMTT.2011.2178427.
- [2] J. Dickmann et al., "Making Bertha see even more: Radar contribution," *IEEE Access*, vol. 3, pp. 1233–1247, July 2015. doi: 10.1109/ACCESS.2015.2454533.
- [3] A. Schiessl, S. S. Ahmed, A. Genghammer, and L. Schmidt, "A technology demonstrator for a 0.5m × 0.5m fully electronic digital beamforming mm-Wave imaging system," in *Proc. 5th European Conf. Antennas and Propagation (EUCAP)*, Apr. 2011, pp. 2606–2609.
- [4] S. Ayhan et al., "Millimeter-wave radar sensor for snow height measurements," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 854–861, Feb. 2017. doi: 10.1109/TGRS.2016.2616441.
- [5] W. Menzel, "Millimeter-wave radar for civil applications," in *Proc. 7th European Radar Conf.*, Sept. 2010, pp. 89–92.
- [6] M. Geiger, M. Hitzler, S. Saulig, J. Iberle, P. Hügl, and C. Waldschmidt, "A 160-GHz radar with flexible antenna used as a sniffer probe," *IEEE Sensors J.*, vol. 17, no. 16, pp. 5104–5111, Aug. 2017. doi: 10.1109/JSEN.2017.2718100.
- [7] A. Ascher, R. Lenz, and S. Heuel, "A novel approach to measure the automotive radar sensor's robustness against interferers in the lab with realistic scenarios," in *Proc. 19th Int. Radar Symp. (IRS)*, June 2018, pp. 1–9. doi: 10.23919/IRS.2018.8448271.
- [8] S. B. J. Gowdu, M. Asghar, J. Nagel, M. Roßmann, R. Stephan, and M. A. Hein, "Comparison of real and simulated automotive radar targets measured

- with a reference radar," in *Proc. Int. Conf. Electromagnetics Advanced Applications (ICEAA)*, Sept. 2018, pp. 468–471. doi: 10.1109/ICEAA.2018.8520431.
- [9] R. Feger, C. Wagner, S. Schuster, S. Scheiblhofer, H. Jäger, and A. Stelzer, "A 77-GHz FMCW MIMO radar based on an SiGe single-chip transceiver," *IEEE Trans. Microw. Theory Techn.*, vol. 57, no. 5, pp. 1020–1035, May 2009. doi: 10.1109/TMTT.2009.2017254.
- [10] S. Lutz, T. Walter, and R. Weigel, "Lens-based 77 GHz MIMO radar for angular estimation in multitarget environments," *Int. J. Microw. Wireless Technol.*, vol. 6, no. 3–4, pp. 397–404, Apr. 2014. doi: 10.1017/S1579078714000506.
- [11] P. Hügl, F. Roos, M. Schartel, M. Geiger, and C. Waldschmidt, "Radar taking off: New capabilities for UAVs," *IEEE Microw. Mag.*, vol. 19, no. 7, pp. 43–53, Nov. 2018. doi: 10.1109/MMM.2018.2862558.
- [12] L. Godara, "Application of antenna arrays to mobile communications. Part II. Beam-forming and direction-of-arrival considerations," *Proc. IEEE*, vol. 85, no. 8, pp. 1195–1245, 1997. doi: 10.1109/5.622504.
- [13] H. Krim and M. Viberg, "Two decades of array signal processing research: The parametric approach," *IEEE Signal Process. Mag.*, vol. 13, no. 4, pp. 67–94, July 1996. doi: 10.1109/79.526899.
- [14] V. Winkler, "Range Doppler detection for automotive FMCW radars," in *Proc. European Radar Conf.*, Oct. 2007, pp. 166–169. doi: 10.1109/EUMC.2007.4405477.
- [15] F. J. Harris, "On the use of windows for harmonic analysis with the discrete Fourier transform," *Proc. IEEE*, vol. 66, no. 1, pp. 51–83, Jan. 1978. doi: 10.1109/PROC.1978.10837.
- [16] G. Payá-Vayá and H. Blume, *Towards a Common Software/Hardware Methodology for Future Advanced Driver Assistance Systems (River Publishers Series in Transport Technology)*. Gistrup, Denmark: River Publishers, 2017.
- [17] H. Rohling, "Radar CFAR thresholding in clutter and multiple target situations," *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES-19, no. 4, pp. 608–621, July 1983. doi: 10.1109/TAES.1983.309350.
- [18] M. Harter, J. Hildebrandt, A. Ziroff, and T. Zwick, "Self-calibration of a 3-D-digital beamforming radar system for automotive applications with installation behind automotive covers," *IEEE Trans. Microw. Theory Techn.*, vol. 64, no. 9, pp. 2994–3000, Sept. 2016. doi: 10.1109/TMTT.2016.2593731.
- [19] J. Li and P. Stoica, *MIMO Radar Signal Processing*. New York: Wiley-IEEE Press, 2008.
- [20] I. Bekkerman and J. Tabrikian, "Target detection and localization using MIMO radars and sonars," *IEEE Trans. Signal Process.*, vol. 54, no. 10, pp. 3873–3883, Oct. 2006. doi: 10.1109/TSP.2006.879267.
- [21] T. E. Tuncer and B. Friedlander, Eds., *Classical and Modern Direction-of-Arrival Estimation*. Amsterdam, The Netherlands: Elsevier, 2009. [Online]. Available: <https://www.sciencedirect.com/book/9780123745248/>
- [22] D. Brandwood, *Fourier Transforms in Radar and Signal Processing (Artech House Radar Library (Hardcover))*. Norwood, MA: Artech House, 2003.
- [23] C. A. Balanis, *Antenna Theory: Analysis and Design*, 2nd ed. New York: Wiley, 1996.
- [24] H. J. Visser, *Array and Phased Array Antenna Basics*. Hoboken, NJ: Wiley, 2005.
- [25] M. Eric, A. Zejak, and M. Obradovic, "Ambiguity characterization of arbitrary antenna array: Type I ambiguity," in *IEEE 5th Int. Symp. Spread Spectrum Techniques Applications-Proc. Spread Technology Africa*, Sept. 1998, vol. 2, pp. 399–403. doi: 10.1109/ISSSTA.1998.723814.
- [26] M. C. Sullivan, *Practical Array Processing*. New York: McGraw-Hill, 2008.
- [27] C. Vasanelli, R. Batra, A. Di Serio, F. Boegelsack, and C. Waldschmidt, "Assessment of a millimeter-wave antenna system for MIMO radar applications," *IEEE Antennas Wireless Propag. Lett.*, vol. 16, pp. 1261–1264, Nov. 22, 2016. doi: 10.1109/LAWP.2016.2631889.
- [28] R. C. Johnson, *Antenna Engineering Handbook*, 3rd ed. New York: McGraw-Hill, 1993.
- [29] R. Schmidt, "Multiple emitter location and signal parameter estimation," *IEEE Trans. Antennas Propag.*, vol. 34, no. 3, pp. 276–280, Mar. 1986. doi: 10.1109/TAP.1986.1143830.
- [30] P. Häcker and B. Yang, "Mixed-order and higher-order ambiguity functions," in *Proc. IEEE Intelligent Vehicles Symp. (IV)*, June 2011, pp. 752–757. doi: 10.1109/IVS.2011.5940578.
- [31] C. Schmid, R. Feger, C. Pfeffer, and A. Stelzer, "Motion compensation and efficient array design for TDMA FMCW MIMO radar systems," in *Proc. 6th European Conf. Antennas and Propagation (EUCAP)*, Mar. 2012, pp. 1746–1750. doi: 10.1109/EuCAP.2012.6206605.
- [32] J. Bechter, F. Roos, and C. Waldschmidt, "Compensation of motion-induced phase errors in TDM MIMO radars," *IEEE Microw. Wireless Compon. Lett.*, vol. 27, no. 12, pp. 1164–1166, Dec. 2017. doi: 10.1109/LMWC.2017.2751301.
- [33] F. Roos et al., "Compressed sensing based single snapshot DoA estimation for sparse MIMO radar arrays," in *Proc. German Microwave Conf. (GeMiC)*, Mar. 2019, pp. 1–4. doi: 10.23919/GEMIC.2019.8698136.

